



3D UHR seismic and back-scattering analysis for seabed and ultra-shallow subsurface classification

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Abstract

Recently, the seabed classification method based on back-scattering data of multi-beam echo-sounder (MBES) is widely used to analyze the distribution of seabed sediment. Although various analysis methods for seabed classification using multi-spectral MBES have been developed, they are limited in securing penetration depth to consider the characteristics of the shallow subsurface structure. In this study, the seabed and ultra-shallow subsurface classification was performed by comparative analysis of box corer sampling, back-scattering, and 2D/3D ultra-high-resolution (UHR) seismic data obtained from Yeongil Bay, South Korea. We proposed a process for seismic ultra-shallow subsurface classification by the segmentation of the primary seabed reflection wavelet and the amplitude analysis. The seabed-reflected amplitude and back-scattering intensity showed similar mapping trends in the relatively homogeneous and thick surface sediment. On the other hand, it was confirmed that back-scattering data and seabed-reflected amplitude show different patterns when the subsurface structure is related to the seabed surface. It is presumed that because seismic data containing relatively low-frequency components have a deeper penetration depth than MBES, they contain more characteristics of the ultra-shallow subsurface than back-scattering data. These were determined that back-scattering has advantages in representing acoustic anomaly distribution by surface sediment type, and seabed-reflected amplitude is advantageous for representing sediment type by ultra-shallow subsurface. In particular, these results were well shown when the surface sediment thinly covered the rocky bottom. Therefore, it is necessary not only to analyze the back-scattering of MBES but also the ultra-shallow subsurface features through seismic data for valid seabed classification.

Keywords Seabed classification · Ultra-shallow subsurface structure · 2D/3D UHR seismic · MBES back-scattering

Introduction

As marine activities industrial operations expand into the ocean, the coastal site conditions must be accurately identified. In particular, coastal areas are more accessible for development compared to offshore, more accurate analysis of seabed and subsurface characteristics is required because marine environmental changes occur frequently.

The Methods for identifying the geological characteristics of seabed and subsurface can be categorized into direct and indirect approaches. Representative direct methods, such as coring and seabed sampling, typically involve analyzing obtained seabed sediment samples to determine seabed sediment characteristics (Barbagelata et al. 1991). While method can determine accurate seabed information in the sampled area, and due to lower efficiency and cost-effectiveness, it is mainly used as verification data for indirect methods.

Various studies have been conducted on indirect methods-based seabed classification to identify the seabed sediment types and acoustic responses (LeBlanc et al. 1992; Schock et al. 1992; Preston 2009; Lied et al. 2004; Brown et al. 2011; Zhang et al. 2023), sub-bottom imaging techniques using echo-sounder and side-scan sonar (SSS) have been mainly used (Tamsett 1993; Smith and Li 1966; Collier and Brown 2005; Subarsyah et al. 2021; Khomsin et al. 2023). In particular, because back-scattering data through MBES

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reflects the influence of grain size and texture of the seabed sediment materials, research related to seabed classification using back-scattering data has been actively conducted recently (Prager et al. 1995; Mair et al. 2012; Lamarche and Lurton 2018; Trzcinska et al. 2020; Ji et al. 2020; Viala et al. 2021; Tang et al. 2022). However, because the MBES for coastal areas uses sound sources in the 100–400 kHz frequency band, sediment information is limited to the seabed surface. To overcome these limitations, recent efforts have been to perform seabed classification and characterization using multi-spectral MBES (Clarke 2015; Brown et al. 2019; Costa 2019; Fezzani et al. 2021; Menandro et al. 2023; Ntouskos et al. 2023; Syariz and Fauzy 2023; Bai et al. 2023). Multi-spectral back-scattering data allows for the scattering characterization of frequency-dependent of seabed sediment, providing the advantage of influencing penetration depth. However, MBES data are dependent on seabed sediment type and properties, and it does not reflect the sediment information within a few meters of the seabed. Therefore, additional engineering data are required for seabed and ultra-shallow subsurface analysis.

MBES can provide high-resolution seabed information due to its operating frequency, but its low penetration depth mainly provides engineering information about the seabed surface. Therefore, marine seismic surveys are also commonly utilized to determine the subsurface geological characteristics. Marine seismic surveys using sources with several to hundreds of Hz frequency range are typically conducted for oil and gas exploration and deep geological structure analysis, shallow seismic surveys using sources with hundreds of Hz frequency range have been actively conducted for coastal management and engineering purposes (Lee et al. 2004; Duarte et al. 2017). Seismic data have a deeper penetration depth than MBES data, it is useful for identifying subsurface geological structures, and can be used to analyze the characteristics of the seabed and subsurface using seabed reflection wave (Lee et al. 2008; Jaśniewicz et al. 2019).

In marine seismic surveys, sources such as Chirp sub-bottom profiler (SBP), sparker, air gun, boomer, and others are commonly used, and research on seismic seabed classification using Chirp SBP with kHz frequency range has been mainly conducted (Schock et al. 1989; Kim et al. 2002; Stevenson et al. 2002; Chiu et al. 2015; Li et al. 2023). Furthermore, various studies have been conducted to analyze the acoustic characteristics distribution of the seabed by comparing SBP data with side-scan sonar (SSS) and MBES (Alevizos et al. 2018; Fakiris et al. 2018; Nitriansyah and Cahyono 2022). Zheng et al. (2021) estimated seabed reflectivity and echo intensity based on the amplitude information of primary and secondary seafloor reflection signals obtained from 2D seismic data using a sparker source. However, many of these previous studies were conducted using

2D seismic data, it is difficult to understand the continuity of spatial characteristic distribution in detail. Mosher et al. (2006) conducted seabed morphology analysis using 3D seismic data and back-scattering data for the purpose of geo-hazard assessment in deep water, compared the horizontal and vertical resolutions of the two datasets and performed a comparative analysis of back-scattering and reflectivity.

In this study, we conducted the comparative analysis of seabed surface and ultra-shallow sediment characteristics by direct and indirect methods such as 2D/3D seismic, back-scattering, and sampling data. The acquisition of 2D/3D UHR seismic data was conducted in the coastal area near Yeongil Bay, South Korea, using the portable 3D seismic system (KIGAM EOS3D) (Shin et al. 2021). Bathymetry and back-scattering data were obtained through MBES survey. The bathymetry data were used for swell and tidal correction of seismic data, and the back-scattering data were compared and analyzed with seismic seabed reflection data. Also, the box corer and grab sampling were performed to identify actual seabed material. The seismic seabed reflection data were segmented using the seabed detection method based on the gradient vector flow (GVF) proposed by Lim et al. (2024), and seismic seabed classification was performed using the amplitude in the seabed reflection wavelet. From sampling data, the seabed sediment type in the survey area was confirmed to be sandy and silty area, and the presence of rock outcrops was confirmed through back-scattering data. Changes in seabed sediment trend were observed based on the topography of rock outcrops. In particular, in areas where the seabed sediment was thick, seismic and back-scattering data showed similar anomaly distributions. However, in areas where the seabed sediment was thin and covered over rock outcrop, differences in patterns between seismic and back-scattering data were observed. We propose the combined use of seismic and back-scattering data for effective seabed classification mapping by analyzing the seabed and ultra-shallow subsurface through back-scattering and seismic data.

Methodology

In general, MBES is mainly used for seabed mapping based on bathymetry and back-scattering data. On the other hand, seismic surveys are applied to geological feature imaging and physical property model construction through signals reflected or refracted from subsurface structures. They have in common that they record signals from the seabed, but they depict information at different penetration depths due to differences in the frequency bands of the sound sources used. MBES for coastal survey employs source frequencies in the range of several hundred kHz, whereas the frequency range of sources used in shallow marine seismic applications

is typically in the range of several hundred Hz. The MBES back-scattering and the seabed-reflected seismic data represent information within a few to tens of centimeters and several meters below the seabed, respectively. In other words, the resolution is higher in the back-scattering data from the MBES, compared to the seismic reflection wave; while, the penetration depth is deeper for the seismic reflection wave compared to the back-scattering data (Fig. 1). Based on the difference in their resolutions and penetration depths, it can be said that MBES and shallow seismic data represents the characteristics of the seabed and ultra-shallow subsurface, respectively. Therefore, the common approach for seabed classification involves the utilization of back-scattering data. However, when the seabed sediment is related to sub-surface geological features, such as a rocky bottom where sediments are very thinly deposited on the seabed surface, it is difficult to determine seabed sediment type using only back-scattering data. In this study, we utilized not only back-scattering data but also shallow seismic data, which

can provide information within several meters below the seabed for effective seabed classification and analysis from a coastal management and engineering perspective. Additionally, box corer sampling was conducted to verify the real seabed material type. Although it is possible to directly observation of vertical geological changes by coring, it is cost-effective to obtain widely seabed information through MBES and seismic survey.

In this study, we compared and analyzed back-scattering data and 2D/3D UHR seismic data for seabed and ultra-shallow subsurface classification. Since back-scattering data processing applied a general process, detailed explanation was omitted. The workflow of the seismic seabed reflection characteristic analysis method applied in this study is illustrated in Fig. 2. The segmentation of seabed-reflected wavelet is required to analyze the characteristics of seismic seabed reflection. Seabed-reflected segmentation must be preceded by seabed detection in seismic data, we adopted a guideline-based seabed detection method

Fig. 1 Schematic diagram comparing MBES back-scattering and seismic reflection characteristics on the seabed

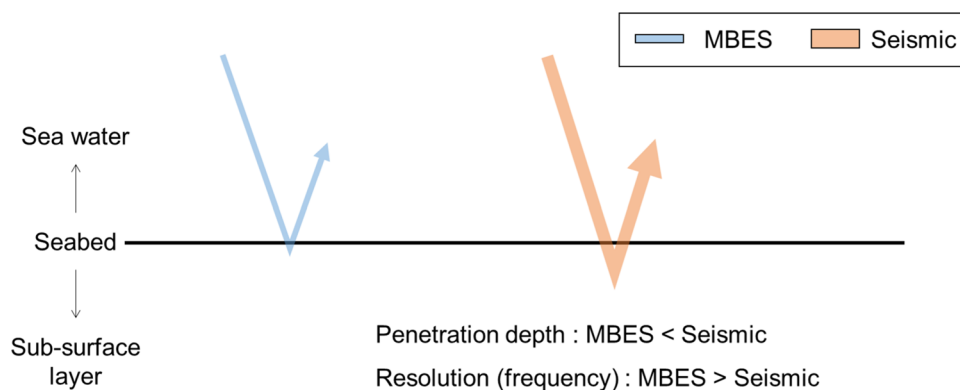
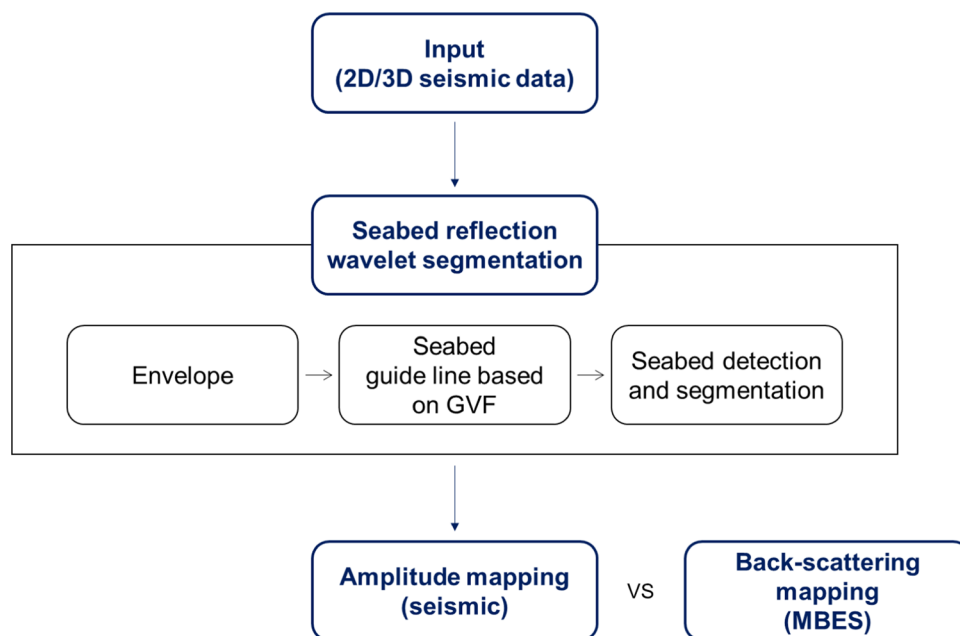


Fig. 2 Workflow of seismic seabed reflection characteristics analysis



derived from the GVF proposed by Lim et al. (2024). This method is known to obtain high accuracy in seabed detection using high-frequency seismic data (Lim et al. 2024). The segmented seabed-reflected wavelet was utilized for amplitude-based analysis of ultra-shallow subsurface classification, and the process of extracting amplitude is shown in Fig. 3. Through wavelet segmentation, we identified t_1 , t_2 , and t_3 and utilized the sum of amplitudes in the positive intervals as a representative amplitude value. Zheng et al. (2021) proposed Eq. (1) to derive the reflection coefficient of primary seabed reflection for seismic data using an omni-directional source:

$$RC = \frac{A_p}{A_d} \times \frac{T_p}{T_d}, \quad (1)$$

where A_d and A_p represent the amplitude of the direct wave and primary seabed reflection, and T_d and T_p represent the first arrival and two-way travel time of the direct wave and primary seabed reflection, respectively. We used near offset gather in 2D seismic data, and in-line data from stack cube data in 3D seismic survey data. However, it is difficult to secure a well-direct wavelet due to the process of 3D stacking. If stable source generation is assumed, the direct wave amplitude and travel time can be omitted. Additionally, the selection of time samples or time intervals to obtain amplitude information of seabed reflection data is user dependent, which limits consistent analysis. Since we can segment the seabed-reflected wavelet, the amplitude information extraction section can be set constant. The representative amplitude value is calculated using Eq. (2):

$$A_s = T_p \sum_{t_1}^{t_2} A_p. \quad (2)$$

A_s represents the seabed-reflected seismic amplitude, it is calculated as the sum of amplitudes in the positive intervals (from t_1 to t_2 in Fig. 3). We mapped the seismic

classification through the process of generating values trace by trace from 2D/3D stack data.

2D seabed and ultra-shallow subsurface classification

2D seismic and multi-beam data were obtained in the northern of Yeongil Bay, South Korea (Fig. 4). The seismic source was Bolt's 10 cu-in air gun, and the streamer was a 16-channel digital streamer with a 3.125 m group interval developed by AWT. The multi-beam back-scattering data were obtained using the Kongsberg EM2040 with 300 kHz.

Figure 5 shows the back-scattering intensity and seabed-reflected seismic amplitude in the 2D seismic section. The back-scattering and seabed-reflected data in the 2D seismic section show similar trends, but show different patterns in some areas (solid and dotted black box in Fig. 5). In the solid black box area, it is observed that the back-scattering data are more sensitive to changes compared to seabed-reflected data (Fig. 5c). This is because MBES uses relatively high frequencies, allowing it to classify seabed material condition and type with high resolution. In the dotted black box area, the thick seabed sediment is observed, transitioning into area related to subsurface geological features (Fig. 5d). The area showing similar trend is presumed to have a relatively homogeneous thick seabed sediment. On the other hand, in area related to subsurface geological features, the seismic section and seabed-reflected amplitude can be used to analysis that the effect is not simply due to the seabed surface condition. If only back-scattering data are utilized, it can be recognized as simply a change in characteristics of seabed surface sediment. But when analyzed together with seismic data, it can be confirmed that it is related to ultra-subsurface structures. Since ultra-shallow subsurface structures can affect seabed classification, back-scattering analysis must be accompanied by seismic section and seabed-reflected amplitude analysis.

Through these results, it is confirmed that conducting analysis on the seabed surface using MBES data and simultaneously considering the ultra-shallow subsurface with

Fig. 3 Schematic diagram comparing MBES back-scattering and seismic reflection characteristics on the seabed

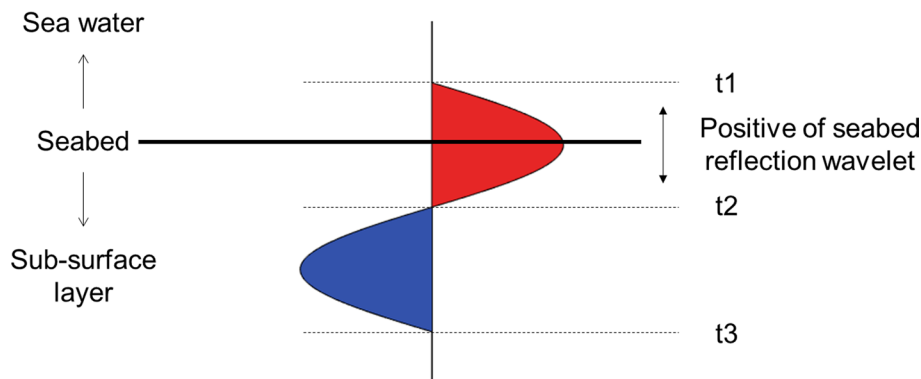
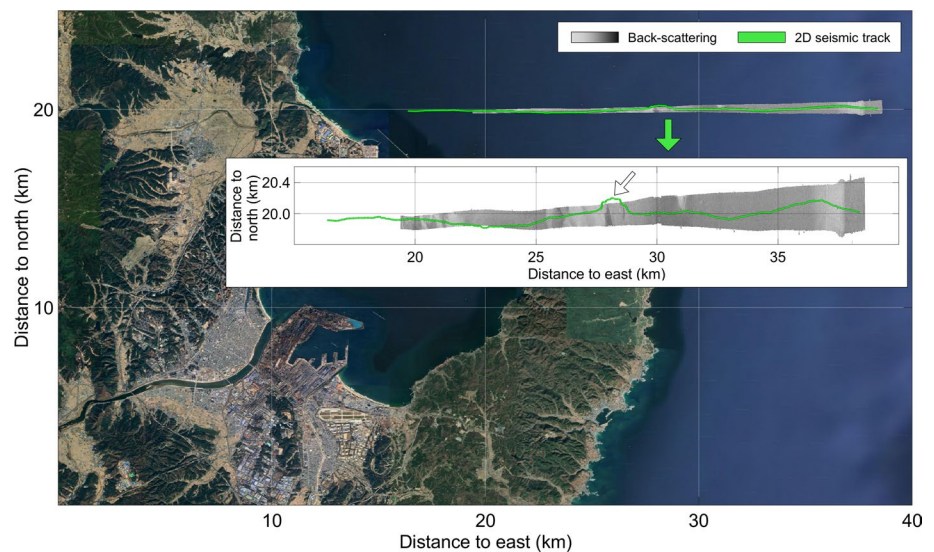


Fig. 4 Schematic diagram comparing MBES back-scattering and 2D seismic survey track. White arrow refers to a non-comparable area with back-scattering data due to off-track seismic survey



seismic data enables valid seabed classification. Additionally, 3D seismic data are essential for spatially continuous seabed and ultra-shallow subsurface classification.

3D UHR seismic survey

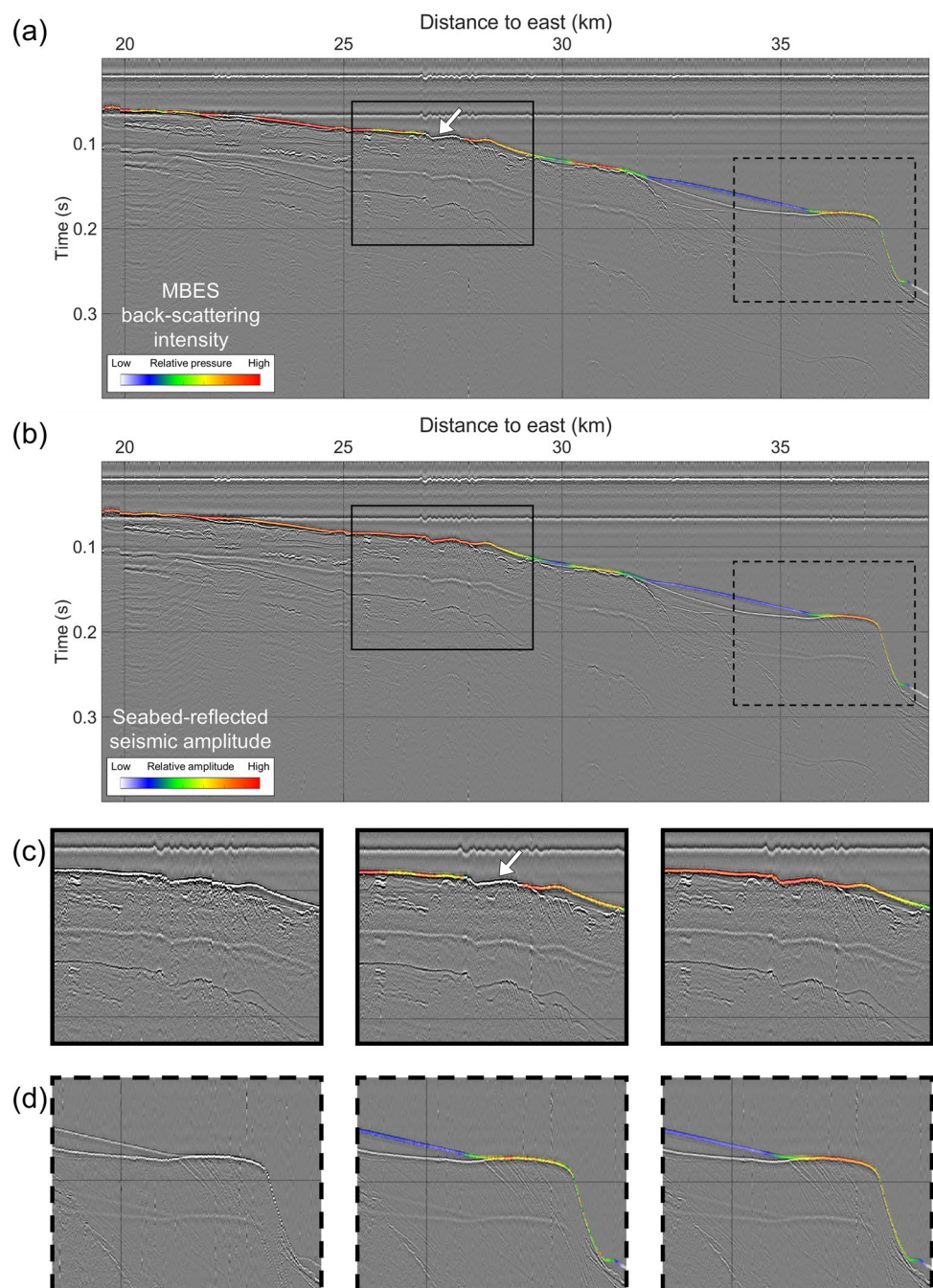
In this study, various marine surveys for seabed and ultra-shallow subsurface analysis were conducted within Yeongil Bay, the southern area of the 2D seismic survey site (Fig. 6). Prior to conducting MBES and seismic surveys, the seabed sediment sampling was performed using the box corer. As a result of the sediment analysis, it was confirmed that Yeongil Bay is divided into areas where sandy and silty seabed sediments are dominant along the ENE-WSW direction (gray dotted line in Fig. 6). The white circles in Fig. 6 indicate locations where seabed material sample recovery failed, and generally occurs when there is rocky bottom. Therefore, in this study, the 3D UHR seismic survey was performed by selecting an area (green box in Fig. 6) that is expected to contain rocky bottom and includes a boundary with ENE-WSW direction.

We utilized the EOS-Streamer system for 3D UHR seismic data acquisition. EOS-Streamer is one of the KIGAM EOS3D systems, representing the 3D UHR seismic system operating 4 arrays of streamers in shallow coastal waters (KIGAM 2017, 2019; Shin et al. 2020, 2021). Data acquisition using the EOS-Streamer was designed as illustrated in Fig. 7. For successful acquisition and processing of 3D seismic survey data, accurate coordinate information of the source and receivers is essential. In this study, broadcast Real-Time Kinematic (RTK) was applied to obtain accurate coordinates for both the source and receivers, ensuring precise positioning information. Broadcast RTK is a network-based RTK service that was commercialized by the

South Korea broadcasting station MBC in 2017. The use of Broadcast RTK allows for real-time RTK correction services with a coverage of 30 km along the coastal areas of South Korea. This provides accurate position information of source and receiver for shallow coastal seismic survey, case of 3D UHR seismic survey conducted in coastal areas utilizing the Broadcast RTK system has been presented (Shin et al. 2024). For efficient launch and recovery of seismic equipment, we applied a wireless communication system using Wi-Fi type for transmitting and receiving each GPS information (green dotted circle in Fig. 7). Additionally, a small deflector and separation rope were used to maintain constant spacing and array configuration of the 4-array streamers (purple hexagon and red line in Fig. 7).

Table 1 presents the data acquisition parameters and 3D seismic volume configuration. Data processing was performed using EOS-Processing (S/W) of EOS3D system. The data processing involves sequential steps, including pre-processing with de-noising, de-bubbling to remove secondary bubbles from the air gun, and swell correction, as illustrated in Fig. 8. In trace-based processing, shot gather is sorted by receiver gather and then processing is performed to improve data quality. In gather-based processing, the same channels of each streamer are grouped, general seismic multi-channel processing is applied, and the 3D stack is performed. Finally, the 3D regularization is applied to the 3D seismic stack cube for improving spatial resolution and connectivity. A description of the processing steps to improve data quality is as follows. In pre-processing, various filtering methods were applied to remove noise from the survey site condition and equipment operation status. For de-bubbling, deterministic deconvolution was performed using the estimated source waveform. And then, the process of performing convolution after removing secondary effects from the estimated source waveform was applied. Sea-bottom detection was performed

Fig. 5 Results of overlaying **a** back-scattering intensity and **b** seabed-reflected seismic amplitude on 2D seismic data. Zoom in results on the **c** solid and **d** dotted black box (left: 2D seismic, center: overlaying the back-scattering on 2D seismic, right: overlaying the seabed-reflected amplitude on 2D seismic). White arrow refers to a non-comparable area with back-scattering data due to off-track seismic survey (in Fig. 4)



through the process shown in Fig. 2, and swell and tidal correction were performed based on MBES bathymetry data. The results of these data processing steps can be visualized in Fig. 9, which includes the fence diagram and time slice. The seabed-reflected seismic amplitude mapping was conducted by trace by trace analysis to the 3D seismic stack cube based on process in Fig. 2.

3D seabed and ultra-shallow subsurface mapping

The MBES bathymetry and backscatter data for the survey area (green box in Fig. 6) were obtained using R2Sonic-2026 with multi-spectral components (200, 300, 400 kHz). For seabed classification, RGB blending processing was applied to the multi-spectral components of back-scattering dataset. These MBES dataset and the seismic seabed-reflected amplitude map derived from the 3D seismic stack cube are shown in Fig. 10. In the bathymetry (Fig. 10a) and back-scattering

Fig. 6 Distribution of seabed materials and the 3D UHR seismic survey area in Yeongil Bay

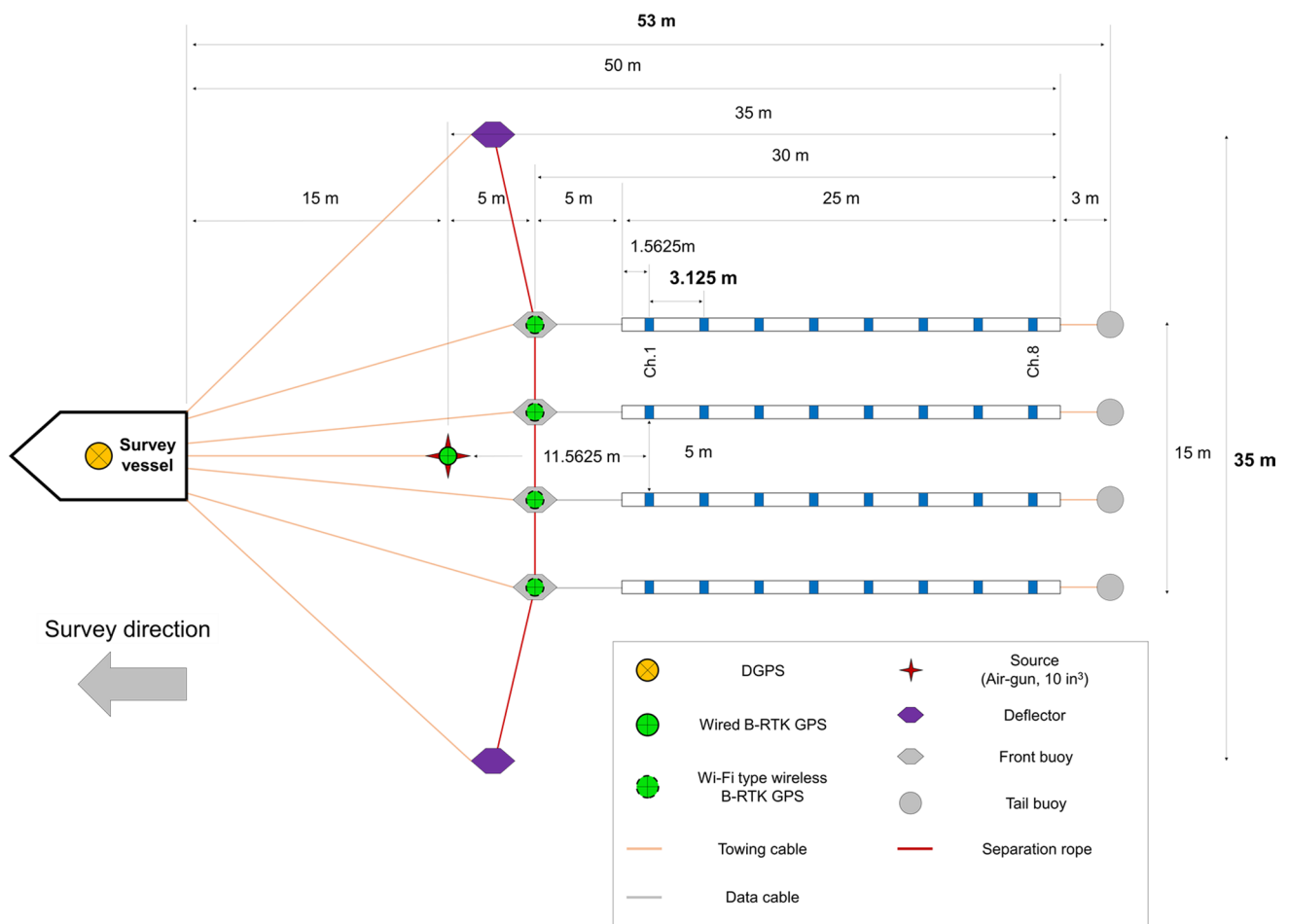
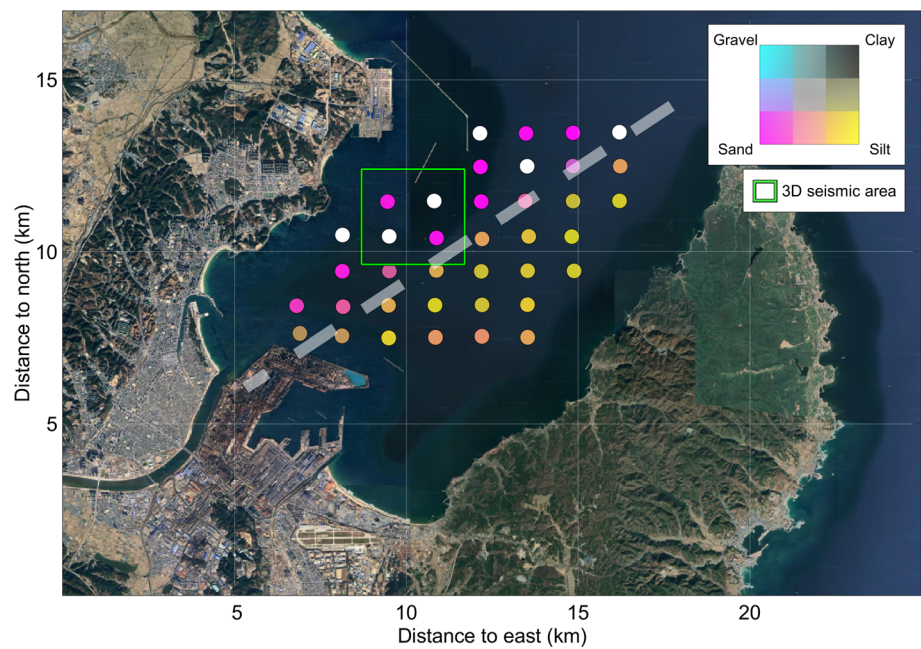


Fig. 7 Schematic diagram of the operational concept for 3D UHR seismic data acquisition using the EOS-Streamer system

Table 1 3D UHR seismic data acquisition and volume set-up parameters

Site area	Yeongil bay, South Korea	Source type (trigger interval)	10 cu-in air gun (2 s)
Coverage area	3.0 km×2.8 km	Number of streamers	4
Bin size	2.5 m	Channels per streamer	8
3D cube size (X×Y×T)	1200×1120×3750	Streamer separation	5 m
Dominant frequency	250 Hz	Group interval	3.125 m

Fig. 8 3D seismic data processing workflow

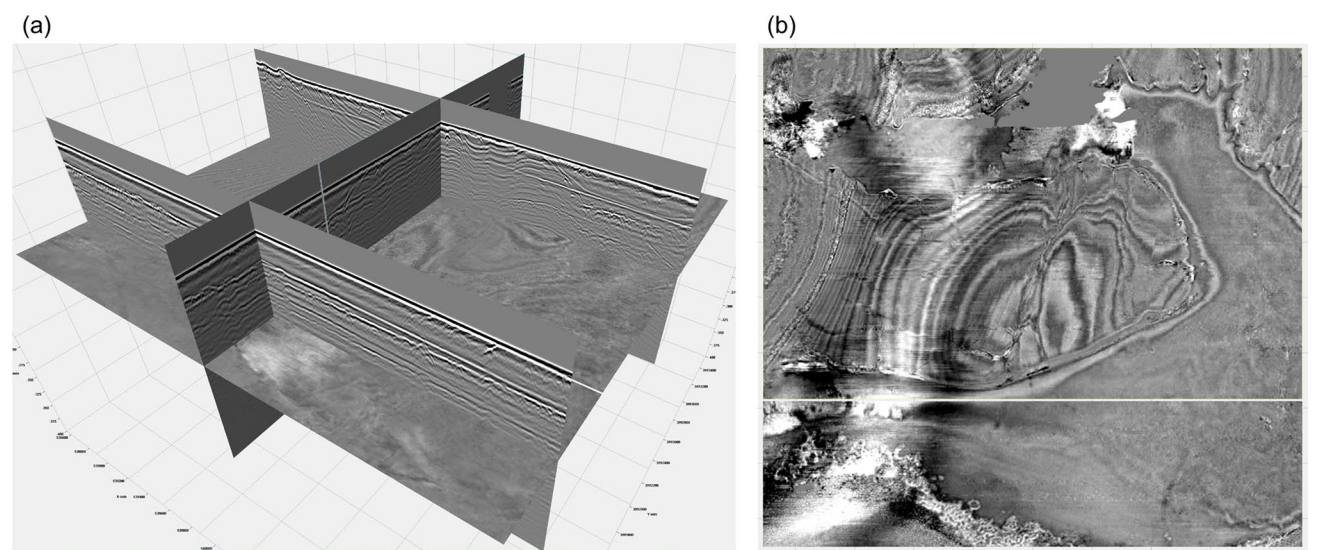
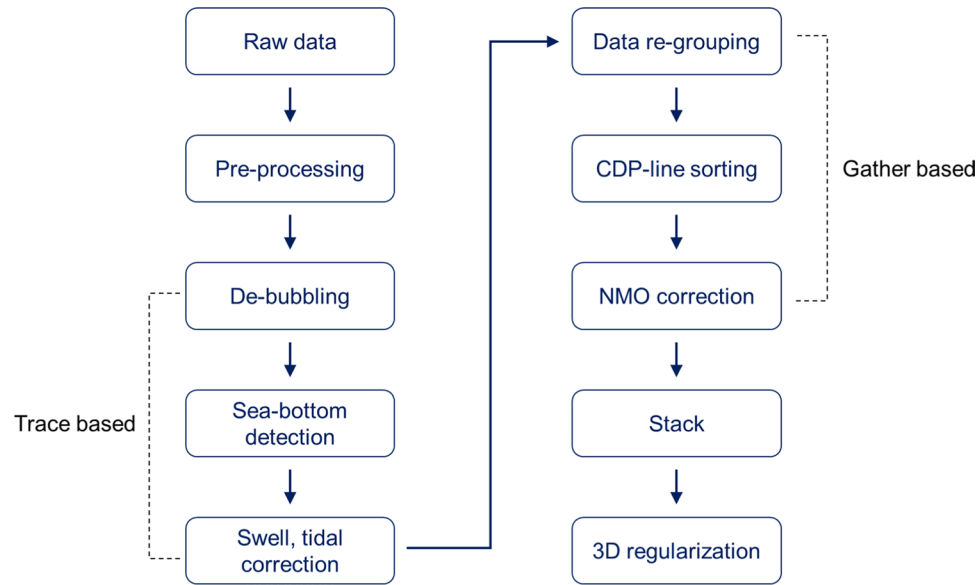


Fig. 9 **a** Fence diagram of 3D stack cube, **b** time slice at 55 ms

(Fig. 10b) map, rock outcrops are observed in the western and northwestern areas, breakwater and artificial structures are confirmed in the central-northern part of the surveyed area. Additionally, it can be confirmed that the distribution

of rock outcrops is clearly visible in the back-scattering data, and that seabed classification is being performed by distinguishing various anomalies on the seabed.

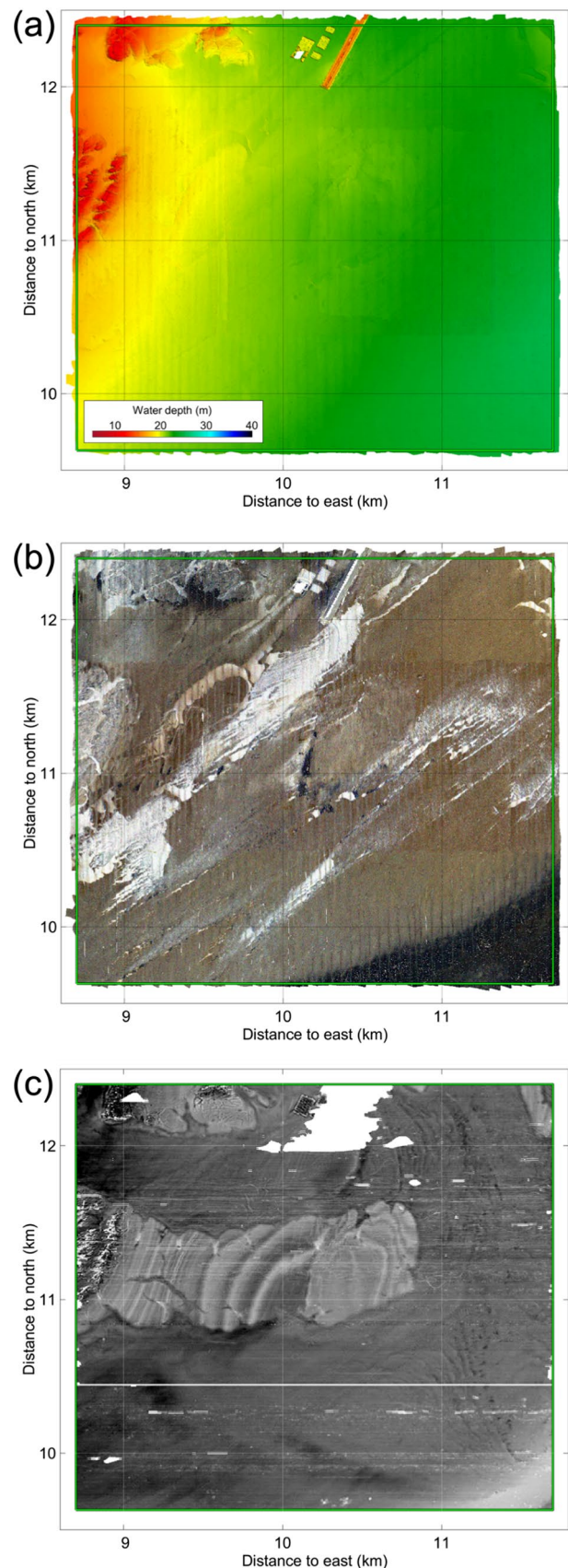
Fig. 10 **a** Bathymetry, **b** RGB blended back-scattering and **c** seabed seismic reflection map

While the structure extending from the western rock outcrops is clearly visible in the seismic seabed-reflected amplitude map (Fig. 10c), it is not observed in the bathymetry and back-scattering map (Fig. 10a, b). The reason why the extension of the rock outcrops is confirmed only in seismic data is that the penetration depth of seismic data is deeper than MBES. The dominant frequency of the obtained 3D seismic data is approximately 250 Hz. Assuming a subsurface P-wave velocity of 1,500 m/s and a vertical resolution of wavelength/4, the seabed-reflected amplitude map includes subsurface information within approximately 1.5 m below the seabed. Therefore, in the seabed seismic reflection map, the upper part of the bedrock layer extending from the exposed bedrock in the west is confirmed to be located within approximately 1.5 m below the seafloor. In the backscatter data, it indicates the surface characteristics of the thin overlaying layer, likely representing a shallow substrate covering the bedrock. Therefore, the seabed-reflected amplitude map confirms that the upper part of the rocky bottom extending from the western rock outcrops is located within about 1.5 m below the seafloor, and the backscatter map shows the seabed surface characteristics of a thin layer covering the upper part of the rocky bottom.

In order to determine the detailed actual seabed surface sediment type in the survey area, additional sampling by grab was conducted in addition to the existing legacy sampling data shown in Fig. 6. Overlaying the distribution of seabed surface sediment type on the seabed-reflected amplitude and backscatter maps results in Fig. 11. In Fig. 11, the sampling area is identified as a sandy-dominated area, and the surface deposition covering the rock bottom is assumed to be a thinly sandy layer. In addition, the part showing strong back-scattering anomaly (white color in Fig. 10b) contains some gravel (red circle in Fig. 10b), so it is assumed that the strong anomaly is caused by gravel.

Figure 12 shows the comparison of the distribution of back-scattering intensity and seismic seabed-reflected amplitude. We conducted the analysis for the parts with the continuity of rocky bottom (Line A in Fig. 12a) and southeast of survey area where strong anomalies are observed (black and white color in back-scattering and seabed-reflected amplitude, Line A in Fig. 12b) by overlaying 3D seismic data, back-scattering and seabed-reflected amplitude map for comparison.

Line A represents the part showing subsurface geological features related rocky bottom. The seismic section at Line A extracted from the 3D seismic cube is shown in Fig. 12b. It can be observed that the surface deposition covering the rocky bottom thickens in the eastward direction from A-2 area (cyan dotted line in Fig. 12b). In Fig. 12c,



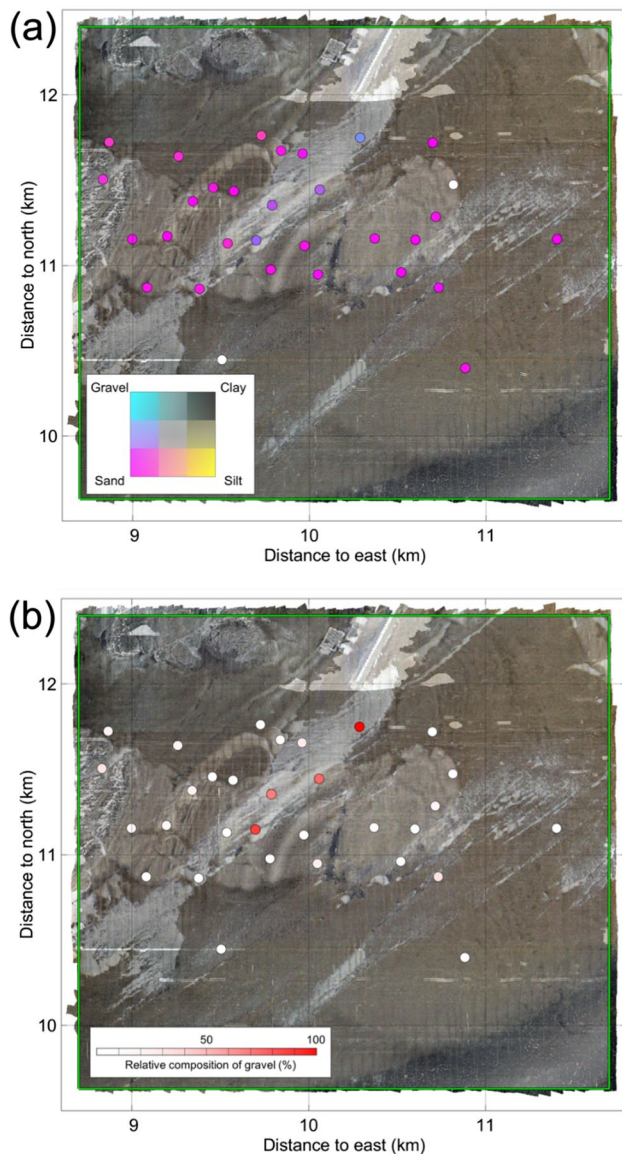


Fig. 11 Overlaying **a** seabed sediment distribution and **b** relative composition of gravel on back-scattering and seabed seismic reflection map

back-scattering data are presumed to show the characteristics of the seabed surface that thinly covers the rocky bottom, and it appears that the seabed is being classified according to the type and composition of seabed surface sediments. On the other hand, the seabed-reflected amplitude map (Fig. 12d) shows amplitude characteristics that include both the rocky bottom and the thin surface layer. There is less amplitude variation in the eastward direction of the A-2 area where the surface deposition covering the rocky bottom thickens. These results indicate that there are limitations in defining seabed sediment through seabed material sampling and back-scattering data analysis only. Therefore, it means that the analysis of ultra-shallow subsurface through seismic

data should be performed together to achieve effective seabed classification.

Line B represents the part where both back-scattering intensity and seabed-reflected amplitude appear strong anomalies. The seismic section at Line B extracted from the 3D seismic cube is shown in Fig. 12e, and it can be observed that the seabed surface deposition layer is pinched out in area B-2 (cyan dotted line in Fig. 12e). The back-scattering intensity and seabed-reflected amplitude in Line B show similar trends overall, and it can be observed that intensity and amplitude become relatively weaker to eastward. In addition, it can be seen that the overall back-scattering data classify the distribution of seabed sediments at a relatively higher resolution than seismic data. This is because when subsurface geological features have low relevance to seabed, back-scattering data acquired using relatively high frequency includes the response to the type and composition of seabed material in high resolution. In the B-2 area, it appears that changes in seabed surface sediment and ultra-shallow subsurface are occurring together. In particular, the both changes in back-scattering intensity and weakening of seabed-reflected amplitude means that seabed surfaced and ultra-shallow subsurface should be compared together.

Conclusions

Recently, seabed classification and surface sediment analysis using back-scattering data have been actively conducted, and in particular, the number of cases considering the various characteristics of seabed surface sediment using multi-spectra MBES is increasing. In this study, we presented a seabed classification and ultra-shallow subsurface analysis process that uses seismic seabed-reflected wave in addition to back-scattering data to improve applicability from the coastal management and engineering purposes.

Seismic survey data have generally been used to image subsurface geological structures and derive property models. We imaged ultra-shallow subsurface features using seismic seabed-reflected waves and performed seabed surface and ultra-shallow subsurface analysis with backscatter data. In order to analyze the characteristics of seabed-reflected waves, segmentation of seabed-reflected wavelets is very important. We applied the GVF-based object boundary extraction concept, one of the image processing techniques, to seabed boundary extraction, and presented a process to identify ultra-shallow subsurface information from the distribution of amplitude values of seabed-reflected wavelets.

The presented seabed surface and ultra-shallow subsurface analysis process was applied to 2D/3D seismic data and back-scattering data obtained at Yeongil Bay, South Korea. In the result of applying 2D seismic data,

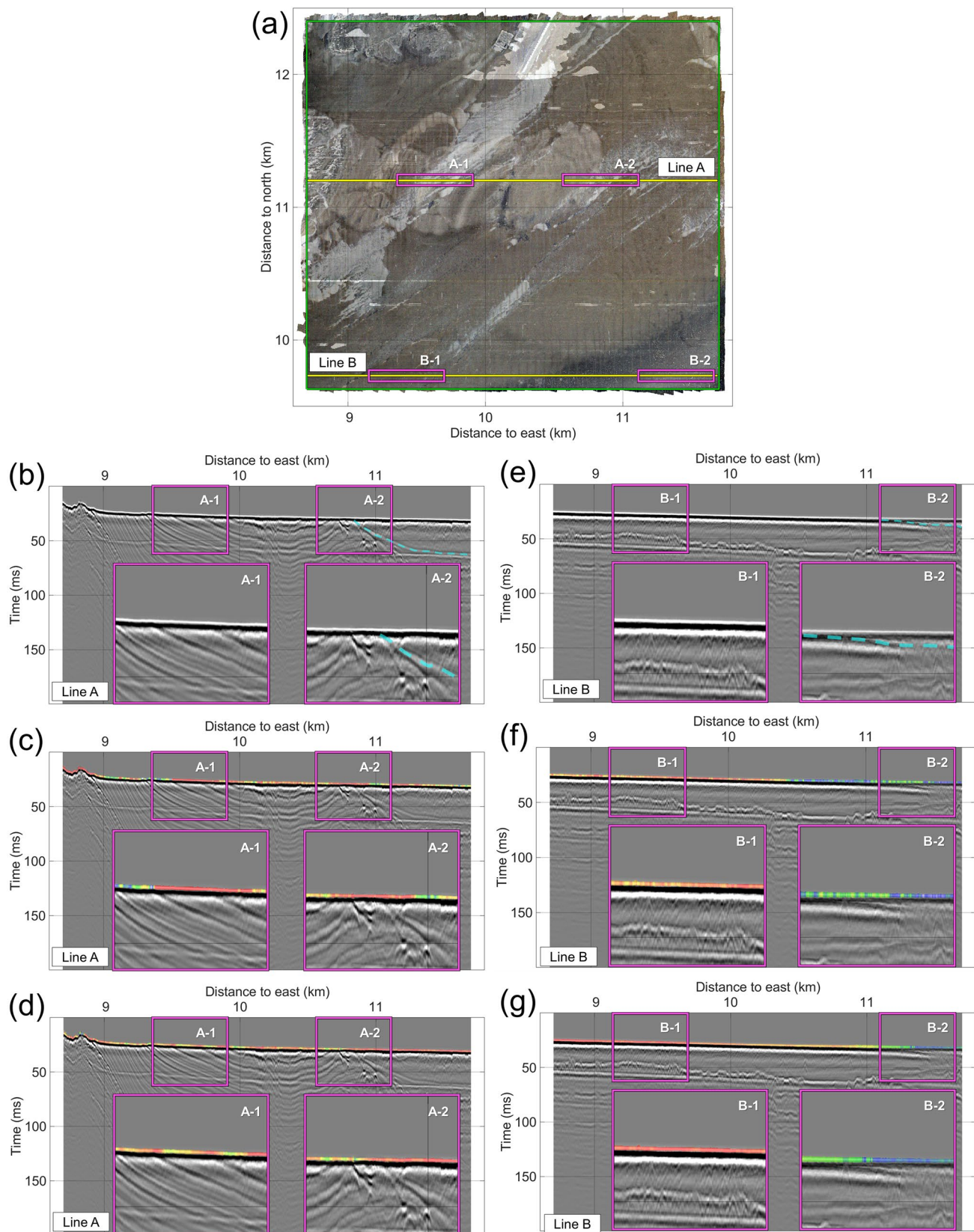


Fig. 12 Comparison of back-scattering and seabed seismic reflection map. **a** Overlaying the back-scattering on the seabed-reflected amplitude map, **b–d** and **e–g** are the seismic section, overlaying the back-scattering and seabed-reflected amplitude on the seismic section at Line A and B

the seabed-reflected amplitude and back-scattering intensity showed similar trends in the relatively homogeneous and thick surface sediment layer. On the other hand, in the seismic section, when the ultra-shallow subsurface structure is related to the seabed surface, it was confirmed that back-scattering data and seabed-reflected amplitude show different patterns. Seismic data containing relatively low-frequency components have a deeper penetration depth than MBES, they contain more characteristics of the ultra-shallow subsurface than back-scattering data.

The southern area of the 2D seismic survey is the area of Yeongil Bay, where the marine industry in Pohang, South Korea is active, and various marine engineering activities are carried out, so information on the seabed and shallow geological structure is important for the purpose of safety marine development. We conducted the analysis of seabed sediment distribution through seabed sampling in the central area of Yeongil Bay. The seabed sediment type was divided into areas dominated by sand and silt based on a specific direction, and the 3D UHR seismic survey was conducted for the divided areas and areas where sampling failed. We performed seabed classification and ultra-shallow subsurface analysis based on 3D seismic and back-scattering data for the efficient utilization of marine spatial information. The 3D UHR seismic survey used the EOS-Streamer option of the KIGAM EOS3D system, and data were obtained for an area of 3.0 km × 2.8 km in the southern of the 2D seismic survey area. Rocky bottom was observed in bathymetry and backscatter data in the survey area, and various anomalies on the seabed surface were observed in backscatter data. In particular, the white band area that appears in the center of the survey area is presumed to be a phenomenon caused by gravel based on comparison with the sampling results. On the other hand, the seabed-reflected amplitude map shows the extension of rock outcrops in the seabed below. In particular, the reason why back-scattering and seabed-reflected amplitudes show different patterns is presumed to be because back-scattering data show only surface sediment thinly covering the rocky bottom; while, seismic data show both rocky bottom and thinly surface sediment layers. In the southeast of the study area, the strong acoustic anomaly is identified in a similar area in the back-scattering and seabed-reflected amplitude map. As a result of comparison with the seismic section extracted from 3D seismic cube, it was confirmed that anomalies appear near the area where the ultra-shallow subsurface layer was pinched out. Since back-scattering data uses the relatively high-frequency band compared to seismic data, it was confirmed that changes in seabed characteristics can be classified at higher resolution.

In the marine industry and engineering, shallow geological information is important in terms of safety, maintenance, and economic efficiency. Therefore, in order to ensure spatial consistency of seabed sediment analysis, seabed sampling,

MBES and seismic survey should be investigated together. Because standard for the vertical depth of surface sediment is relative in classification of seabed type, the spatial information on the seabed surface and ultra-shallow subsurface should be considered together. If ultra-shallow subsurface analysis based on seabed-reflected amplitude is performed considering the frequency characteristics of seismic data, it can be used as useful data for utilizing marine spatial.

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Declarations

Conflict of interest The authors declare that there are no conflicts of interest or competing interests. We certify that the submission is original work and is not under review at any other publication.

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